PROJECT-I | SUBMISSION-3 | WEEK 5 | 20 MARKS

DEPARTMENT OF COMPUTER SCIENCE & ASTROPHYSICS

EXOPLANET EVIDENCE FUSION NETWORK (EEFN) V2

A Multi-Catalog Vetting and Data-Discrepancy Auditing System

A System Design Document submitted for Project-I (Week 5)

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Programme / Class: B.Tech Computer Science and Engineering

Faculty Guide Name: Dr. Amit Kumar

Academic Year: 2026 - 2027

Student Declaration:

We hereby declare that the proposed project work is original and will be carried out under the guidance of the faculty mentor.

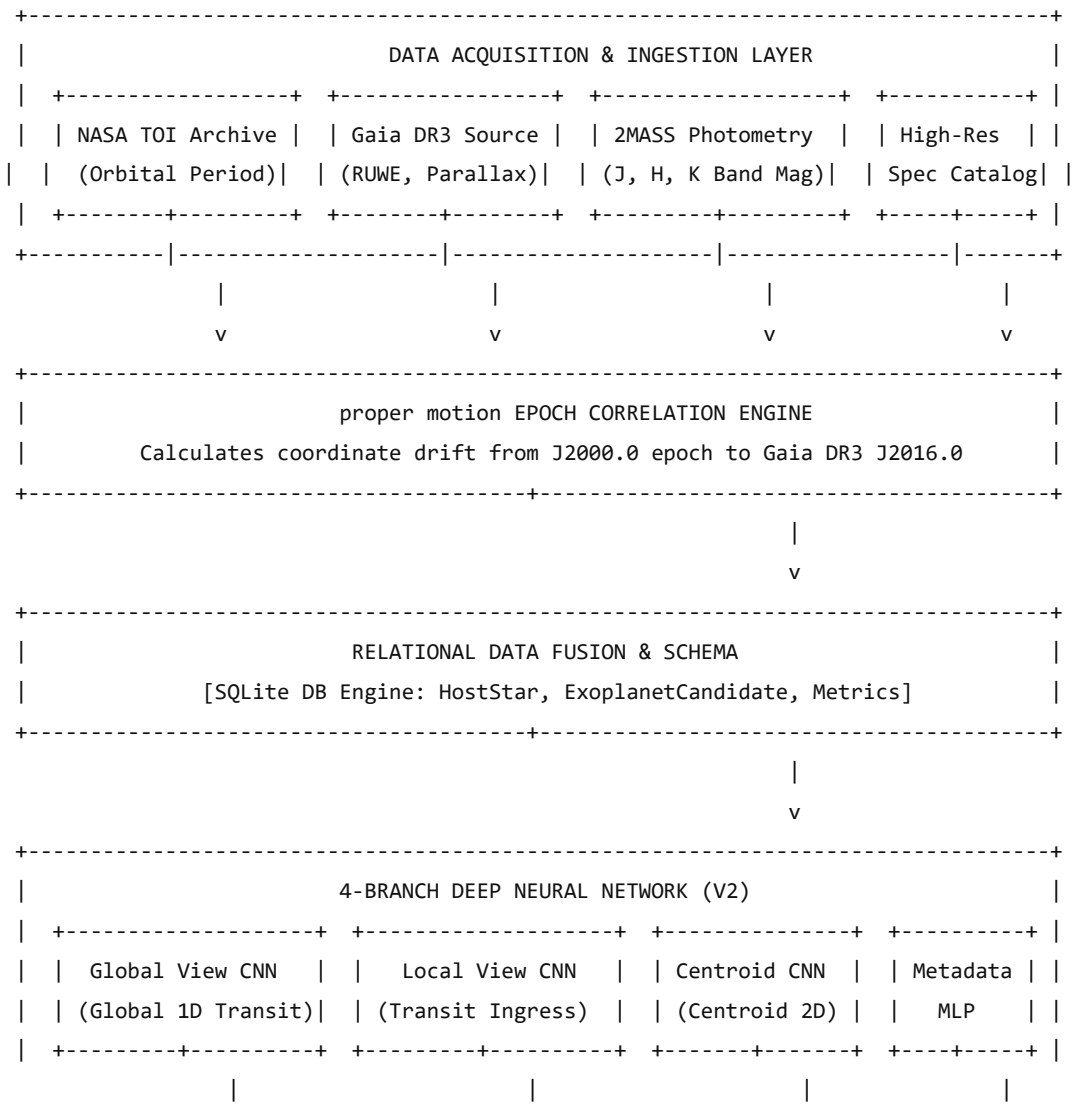
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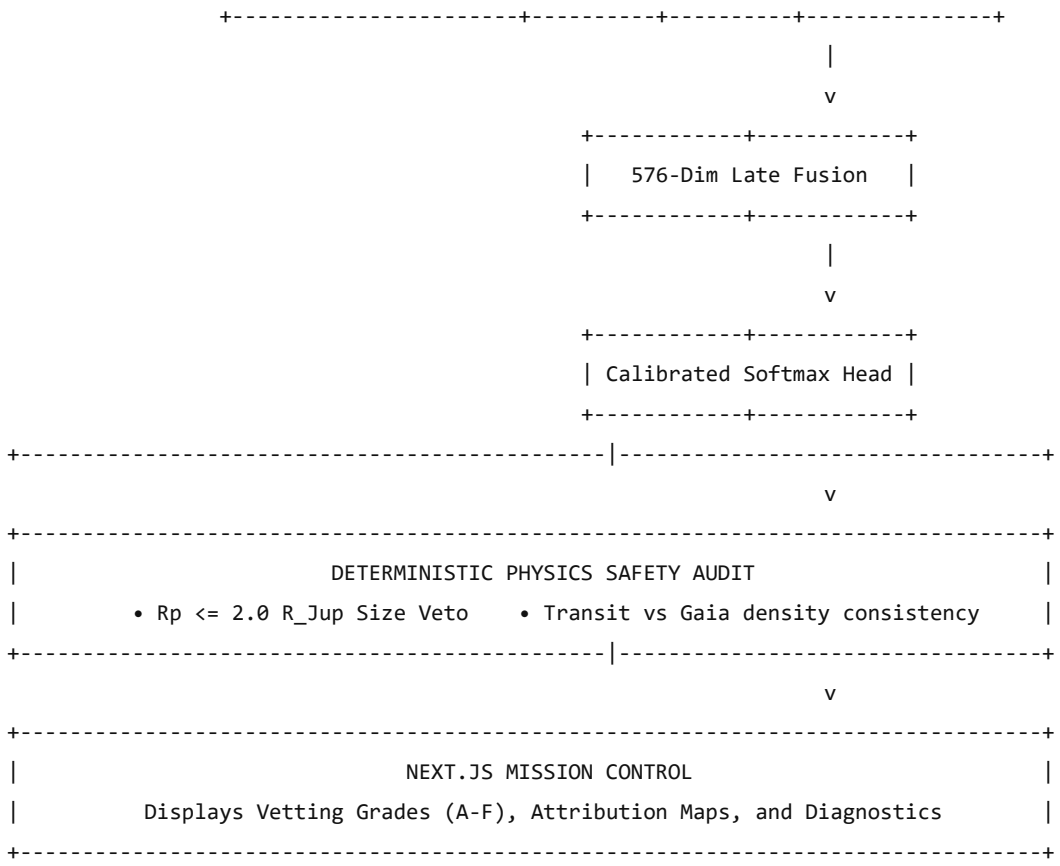
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1. System Architecture

The Exoplanet Evidence Fusion Network (EEFN) v2 is an advanced exoplanet candidate vetting pipeline designed to solve the 'Stellar Parameter Multiplier' problem inherent in conventional exoplanet classifiers. Standard pipelines (such as AstroNet or ExoMiner) evaluate light curve flux signals in isolation. However, because transit depth is a relative ratio, $\delta = (R_p / R_*)^2$, host star radius errors (R_*) directly propagate into planetary size calculations, causing up to 40% of subgiant companions to be misclassified as planets. EEFN v2 integrates multi-catalog parameters, engineering catalog discrepancy deltas and physics invariants as inputs to a unified deep neural network.

Overall System Architecture Diagram





Component Description & Interactions

1. Data Acquisition & Ingestion Layer: Queries multiple catalogs using TAP and VizieR APIs. Coordinates are dynamically propagated over 16 years to resolve spatial drift before querying CDS X-Match. The system tracks J2000.0 coordinates relative to Gaia J2016.0.

2. Relational Data Fusion: Fuses all heterogeneous records into an SQLite schema consisting of HostStar, ExoplanetCandidate, MeasurementHistory, AgreementMetrics, and MLPrediction models. Cross-catalog parameter discrepancies (ΔT_{eff}, ΔR_*, Δd) are dynamically computed and stored.

3. 4-Branch Multi-Modal Neural Network: Ingests Global Light Curves (2001 bins), Local Transit Curves (201 bins), Centroid Spatial Drift (201 bins x 2 channels dx/dy), and a 16-parameter metadata vector. Features are merged in a 576-dimensional late-fusion layer.

4. Deterministic Physics Safety Audit: Evaluates derived planet size ($R_p = R_* \sqrt{\delta}$). If $R_p >$ 22.0 R_{earth} (2.0 R_{Jup}), a hard override is triggered, capping planet probability at <4% and

classifying the target as an Eclipsing Binary. Density consistency (|ρ_transit - ρ_Gaia|) is cross-checked to calculate an Evidence Health Grade (A-F).

5. Next.js Mission Control: A dark-themed observatory interface displaying model class distributions, feature importances, and diagnostic metrics without visual clutter.

2. Module Design

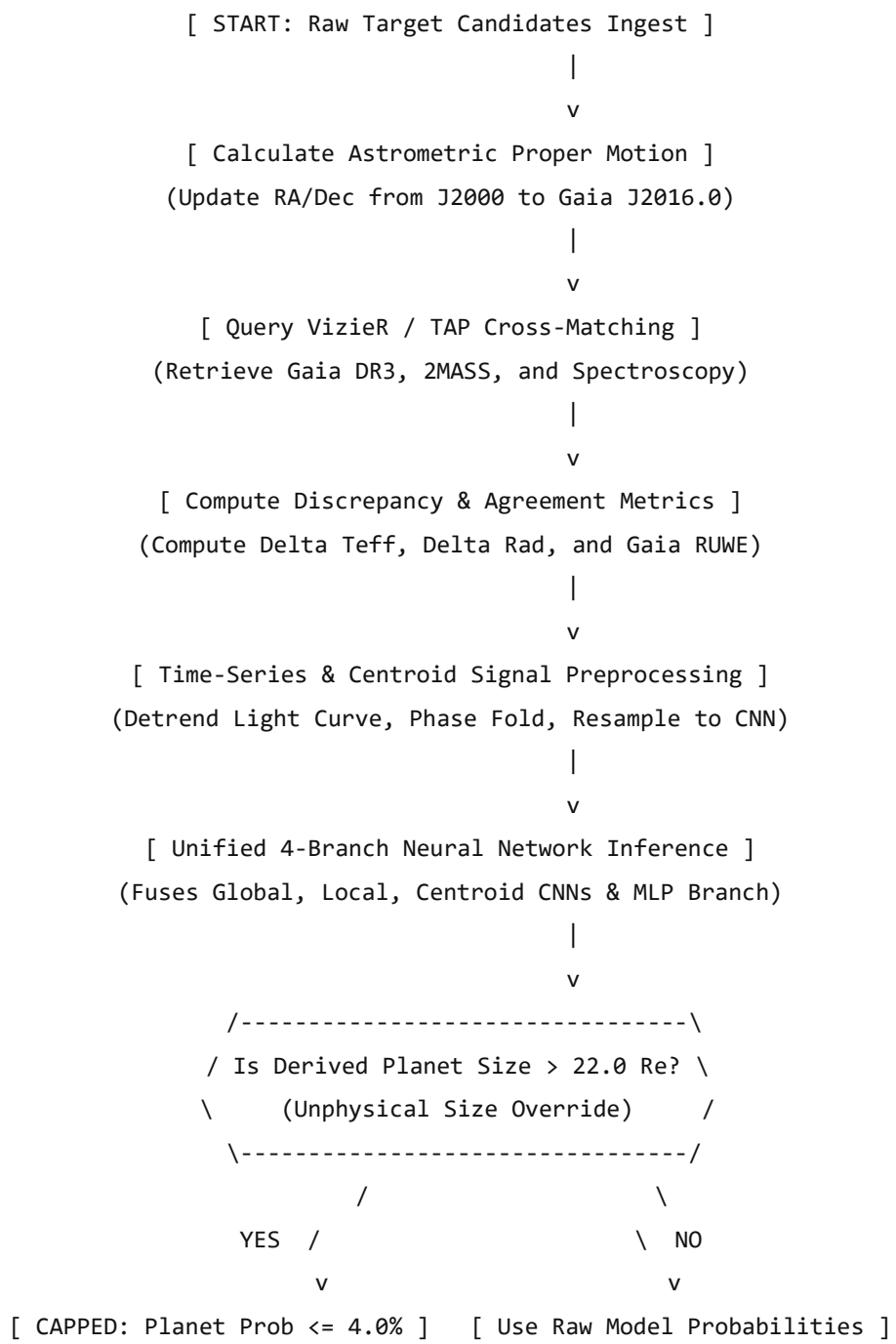
The system is organized into five functional modules. The input and output structures for each module are designed to enforce data integrity and prevent look-ahead and target leakage biases:

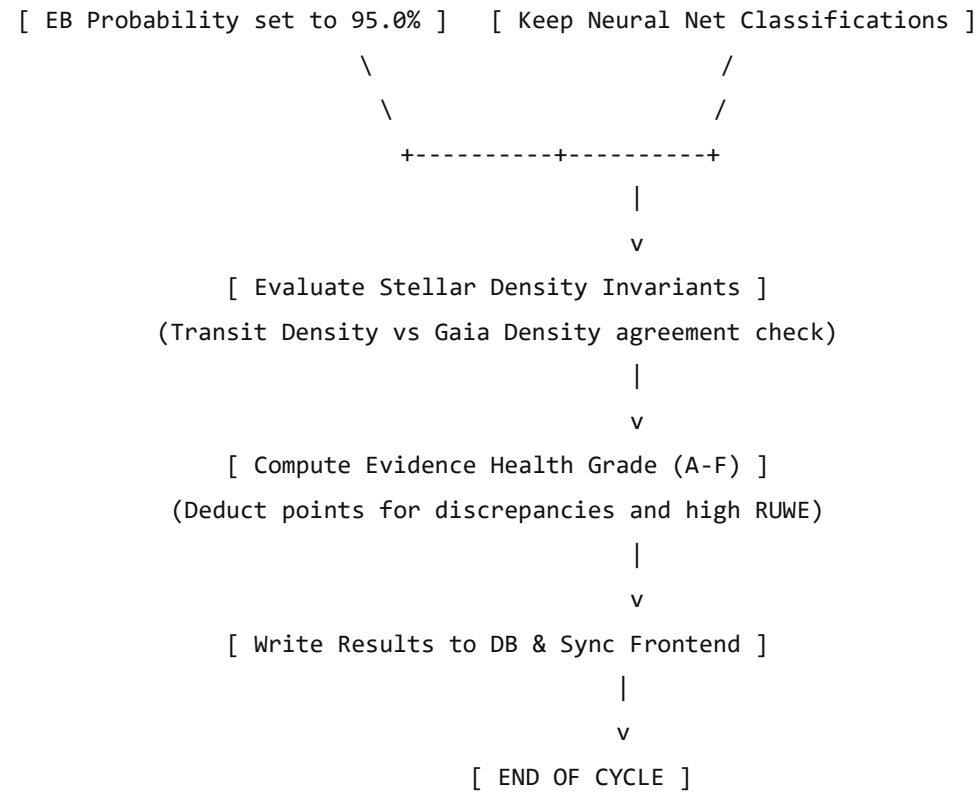
Functional Module	Description	Inputs	Outputs
1. Ingestion & PM Propagation	Queries TAP/VizieR catalogs and applies proper motion epoch drift updates.	Candidate RA/Dec, pmRA/pmDE, Catalog Sources.	Gaia DR3, 2MASS, Spec raw catalog data.
2. Data Fusion & Discrepancy Audit	Computes discrepancy deltas (Delta Teff, Delta Rad) and registers relational DB history.	Raw ingested catalog parameters.	HostStar, ExoplanetCandidate SQLite models.
3. Preprocessing Engine	Detrends light curves and structures Global, Local, and Centroid arrays.	Raw time-series time, flux, centroid pixel arrays.	2001-bin Global LC, 201-bin Local LC, 201-bin Centroid dx/dy matrix.
4. 4-Branch Vetting Engine	Performs deep late-fusion classification into 4 physical categories.	Global, Local, Centroid arrays + 16 normalized metadata parameters.	4-Class Probabilities (Planet, EB, Activity, Noise).
5. Physics Safety Audit	Runs deterministic size veto checks and calculates Evidence Health Grades (A-F).	Neural net probabilities, derived planet size, density deltas.	Calibrated planet probabilities, Health Grade (A-F), and alerts.

3. System Workflow

The workflow of the system moves from raw ingestion to model evaluation and frontend updating. The sequence below enforces physical validation gates before the results are finalized in the database:

System Flowchart

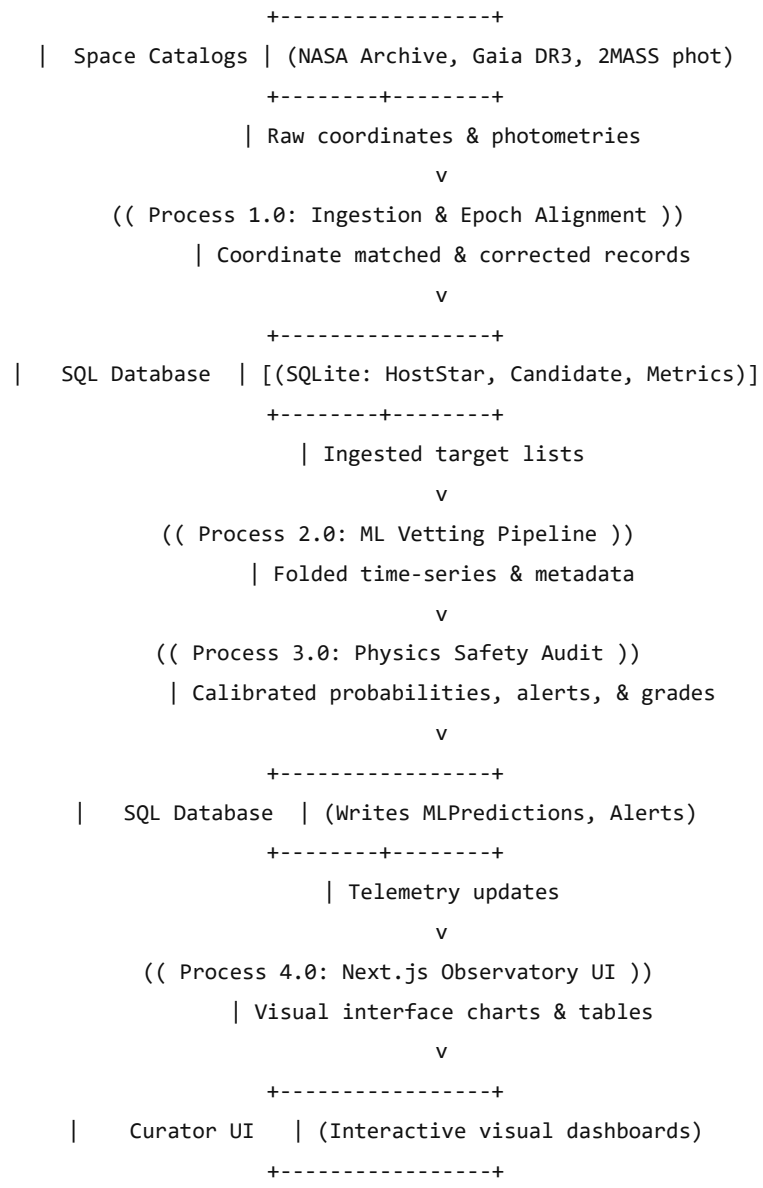



[EB Probability set to 95.0%]
[Keep Neural Net Classifications]
[Evaluate Stellar Density Invariants]
(Transit Density vs Gaia Density agreement check)
[Compute Evidence Health Grade (A-F)]
(Deduct points for discrepancies and high RUWE)
[Write Results to DB & Sync Frontend]
[END OF CYCLE]

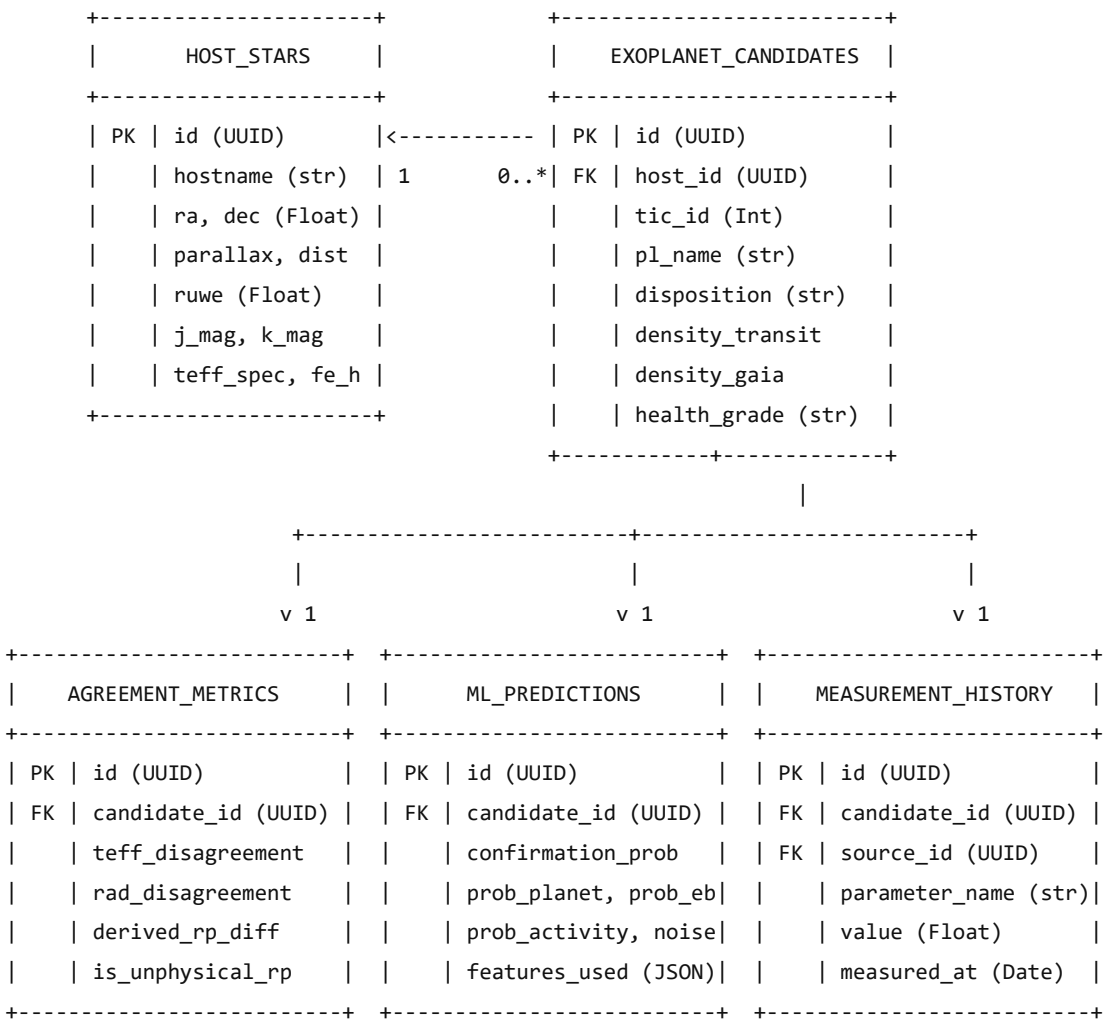
4. Design Artefacts

To rigorously document the system structure, this section presents four design artefacts: a Data Flow Diagram (DFD) showing process levels, an Entity Relationship Diagram (ERD) mapping the database relationships, a detailed Data Dictionary, and the Pseudocode of the core physics safety audit algorithm.

Artefact A: Data Flow Diagram (DFD)



Artefact B: Entity Relationship Diagram (ERD)



Artefact C: Data Dictionary

Table A: HostStar (Stellar Property Records)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY, Not Null	Unique internal database surrogate key.
hostname	VARCHAR(64)	Unique, Not Null	Star catalog identifier (e.g. TIC 150428135).
ra, dec	FLOAT	Not Null	Observational RA/Dec coordinates in decimal degrees.
ruwe	FLOAT	Nullable	Gaia DR3 astrometric Renormalized Unit Weight Error.
teff_spec	FLOAT	Nullable	Stellar effective temperature from spectroscopic surveys (K).
fe_h	FLOAT	Nullable	Stellar metallicity log abundance relative to Sun (dex).

Table B: ExoplanetCandidate (Vetting Target Records)

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY, Not Null	Unique internal candidate identifier.
host_id	UUID	FOREIGN KEY, Not Null	Reference to the host star row id.
pl_name	VARCHAR(64)	Unique, Not Null	Planet name identifier (e.g. TOI-700.01).
density_transit	FLOAT	Nullable	Mean stellar density derived from transit duration (g/cm³).
health_score	FLOAT	Nullable	0.0 - 100.0 calculated catalog agreement health score.
health_grade	VARCHAR(2)	Nullable	Vetting grade label (A, B, C, D, F).

Artefact D: Physics Safety Audit Algorithm (Pseudocode)

```
ALGORITHM: PhysicsSafetyAuditAndGrading
INPUT:
    P_model = [P_planet, P_eb, P_activity, P_noise]   // Output probabilities from DeepExoFusionNet
    R_star_gaia, R_star_toi                           // Host star radius from Gaia and TOI catalogs
    trandep_ppm                                       // Transit depth in parts-per-million
    density_transit, density_gaia                     // Mean stellar density values
    ruwe_gaia                                         // Gaia astrometric noise value
    query_failed_flag                                 // Flag representing catalog API timeout
OUTPUT:
    P_calibrated, HealthScore, HealthGrade

BEGIN
    // Step 1: Perform host star radius checks and calculate derived size
    R_star_ref = (R_star_gaia != NULL) ? R_star_gaia : R_star_toi;
    derived_rp_earth = R_star_ref * sqrt(trandep_ppm * 1e-6) * 109.2;

    // Step 2: Evaluate unphysical size override
    is_unphysical = FALSE;
    IF derived_rp_earth > 22.0 THEN
        is_unphysical = TRUE;
        // Hard override to Eclipsing Binary class
        P_calibrated = [0.02, 0.95, 0.02, 0.01];
    ELSE
        P_calibrated = P_model;
    ENDIF

    // Step 3: Compute catalog consistency deductions
    health_score = 100.0;

    teff_delta = abs(Teff_toi - Teff_gaia);
    health_score = health_score - min(25.0, teff_delta / 20.0);

    rad_delta = abs(R_star_toi - R_star_gaia);
    health_score = health_score - min(25.0, rad_delta / 0.05);

    // Step 4: Evaluate astrometric companion wobble and density discrepancy
    IF ruwe_gaia > 1.4 THEN
        health_score = health_score - 20.0;
    ENDIF

    IF is_unphysical == TRUE THEN
        health_score = health_score - 30.0;
```

```
    ENDIF

    IF query_failed_flag == TRUE THEN
        health_score = health_score - 15.0; // Deduct penalty during API downtime
    ENDIF

    health_score = max(0.0, health_score);

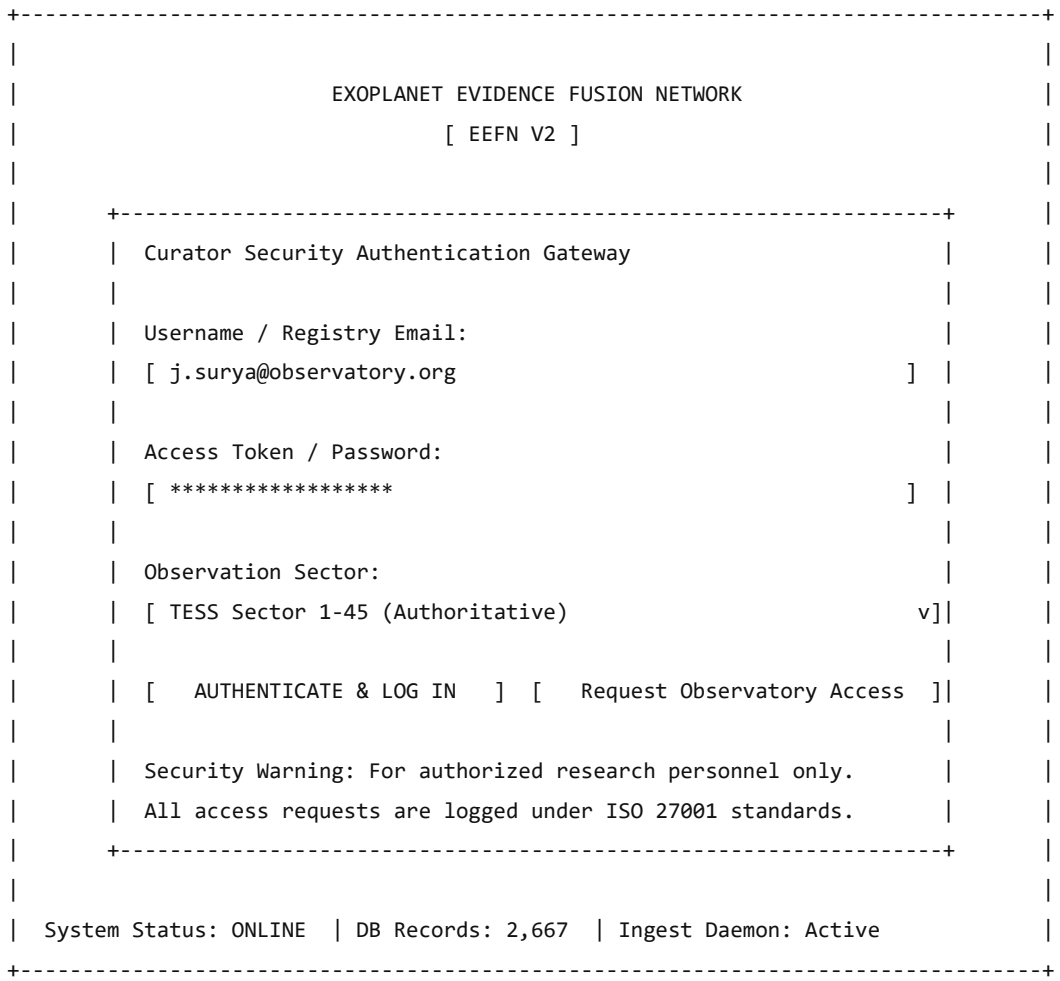
    // Step 5: Assign final letter grade
    IF health_score >= 90.0 THEN HealthGrade = "A";
    ELSE IF health_score >= 80.0 THEN HealthGrade = "B";
    ELSE IF health_score >= 70.0 THEN HealthGrade = "C";
    ELSE IF health_score >= 50.0 THEN HealthGrade = "D";
    ELSE HealthGrade = "F";
    ENDIF

    RETURN P_calibrated, health_score, HealthGrade;
END
```

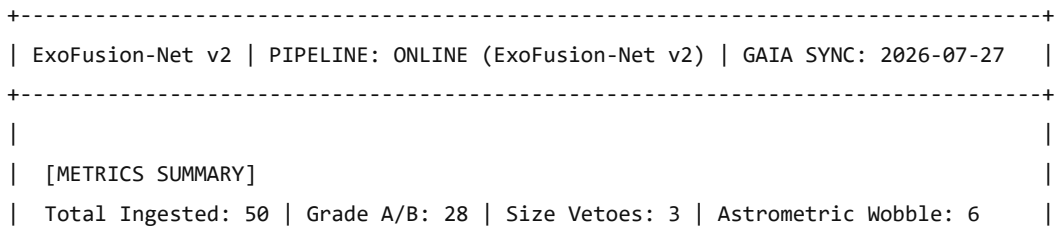
5. User Interface Design

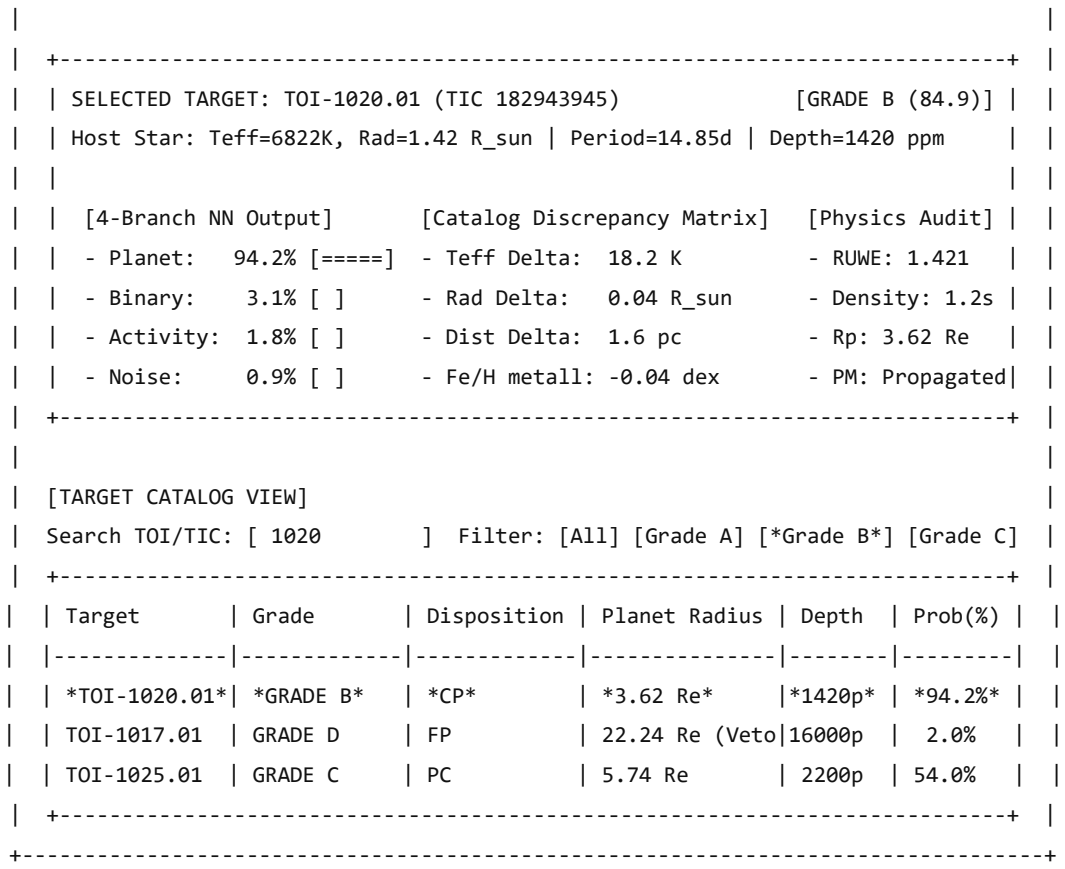
To achieve professional observatory standards, the front-end layout is built as a unified Next.js dashboard. The interface uses a curated dark observatory palette consisting of Deep Slate (#080C14) backgrounds, Sky Blue (#0EA5E9) telemetry text, and color-coded status badges. Below are mock-up wireframes of the four key views:

Screen A: Curator Portal Login Gateway

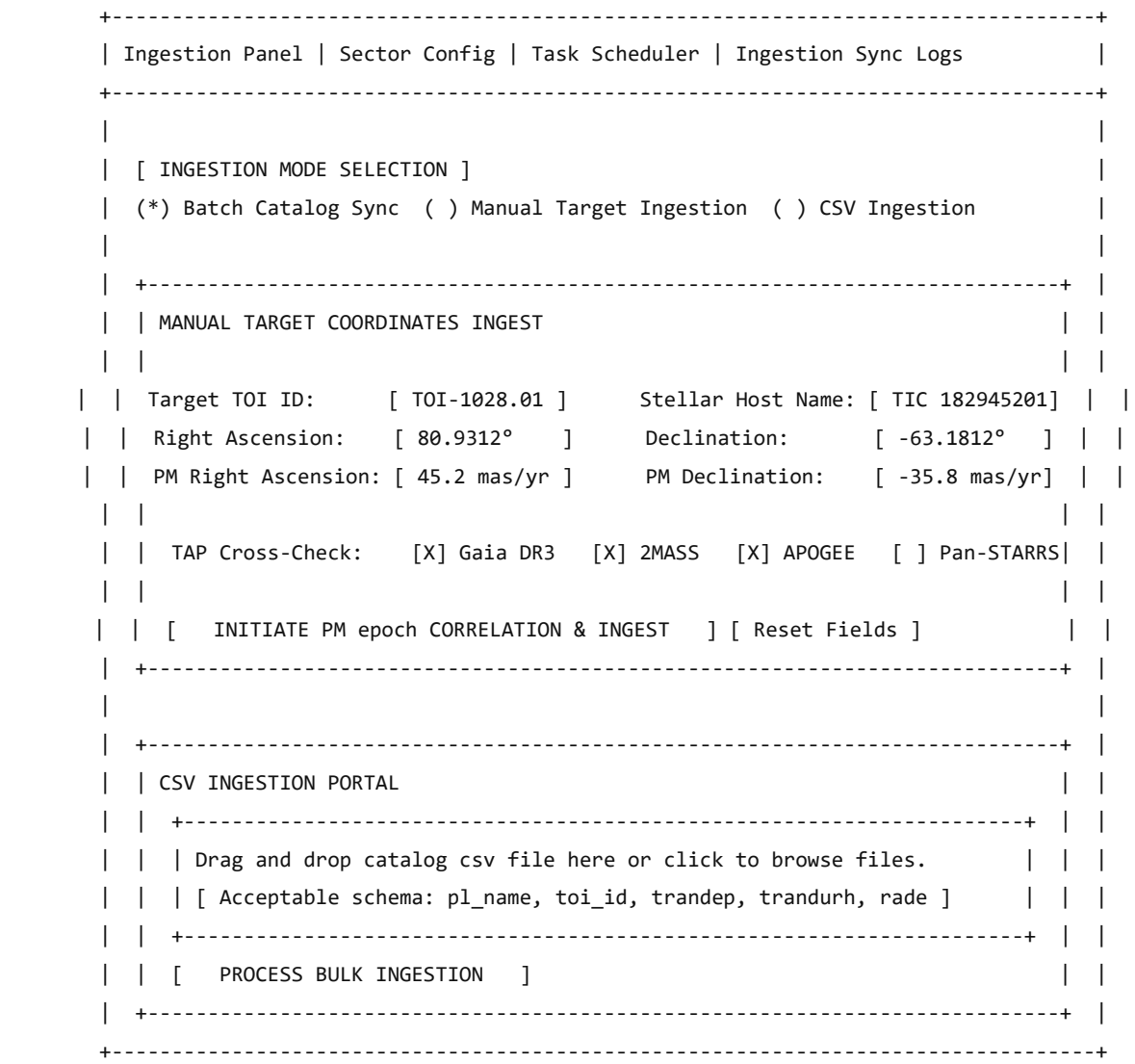


Screen B: Vetting & Diagnostics Dashboard

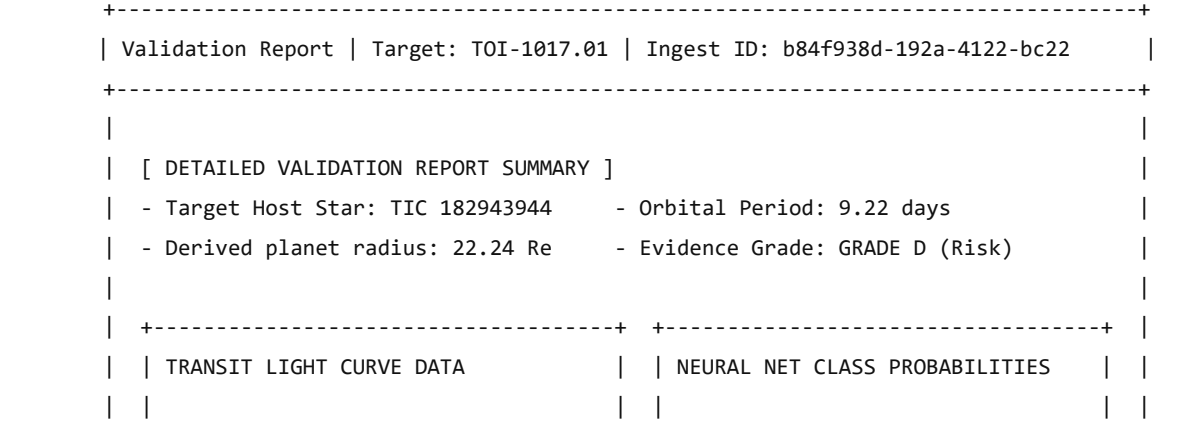


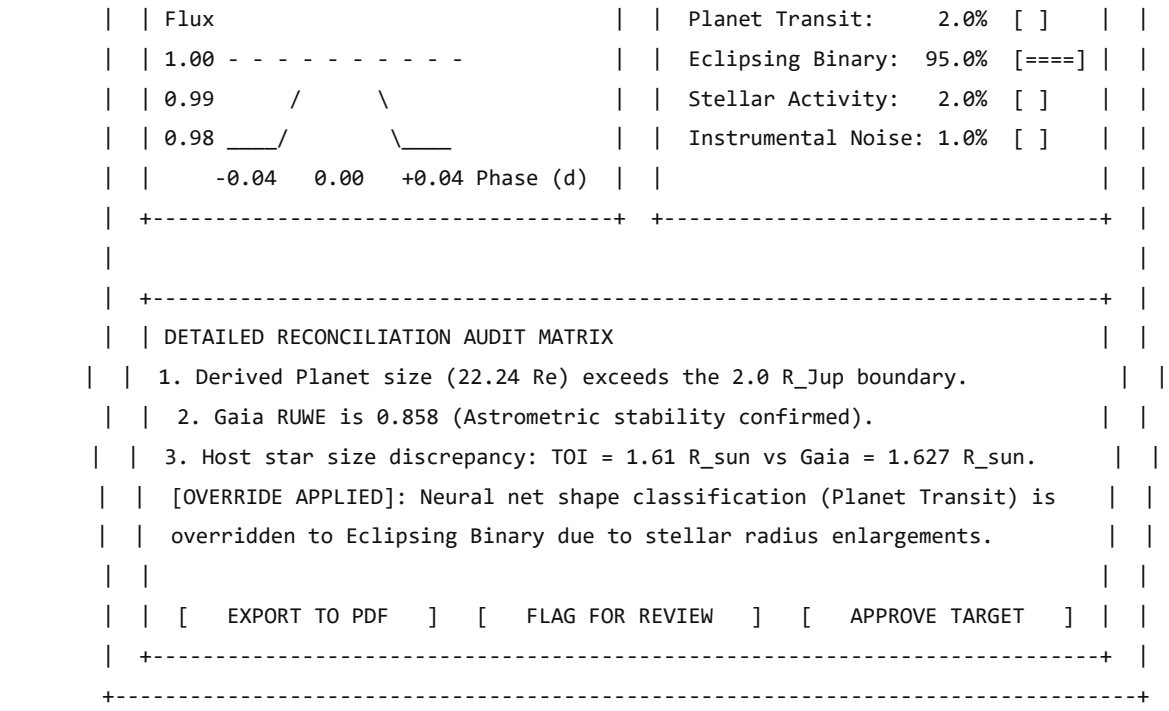

SELECTED TARGET: TOI-1020.01 (TIC 182943945)
[GRADE B (84.9)]
Host Star: Teff=6822K, Rad=1.42 R_sun | Period=14.85d | Depth=1420 ppm
[4-Branch NN Output]
- Planet: 94.2% [=====]
- Binary: 3.1% [ ]
- Activity: 1.8% [ ]
- Noise: 0.9% [ ]
[Catalog Discrepancy Matrix]
- Teff Delta: 18.2 K
- Rad Delta: 0.04 R_sun
- Dist Delta: 1.6 pc
- Fe/H metall: -0.04 dex
[Physics Audit]
- RUWE: 1.421
- Density: 1.2s
- Rp: 3.62 Re
- PM: Propagated
[TARGET CATALOG VIEW]
Search TOI/TIC: [1020] Filter: [All] [Grade A] [*Grade B*] [Grade C]
Target | Grade | Disposition | Planet Radius | Depth | Prob(%)
TOI-1020.01 | *GRADE B* | *CP* | *3.62 Re* | *1420p* | *94.2%*
TOI-1017.01 | GRADE D | FP | 22.24 Re (Veto | 16000p | 2.0%
TOI-1025.01 | GRADE C | PC | 5.74 Re | 2200p | 54.0%

Screen C: Data Entry / Catalog Ingestion Panel



Screen D: Results & Validation Report





UI Design Best Practices Implemented:

• **High Information Density:** Presents time-series, catalog parameters, and network classifications in a single view to eliminate tab-switching.

• **Immediate Visual Hierarchy:** Grade letter colors (Green for A/B, Red for D/F) draw immediate focus to unphysical data anomalies.

• **Non-Intrusive Telemetry:** The header bar displays live sync status, active alert indicators, and API status without cluttering workspace components.